Exploratory Data Analysis of Consumer Complaint Patterns and Company Responsiveness (2020-2025)

Deepali Poudel

DS-620: Data Visualization

Professor Dr. Sharath Kumar Jagannathan

Executive Summary

This report presents an analytical overview of 10,000 sampled records from the Consumer Complaint Database to uncover patterns in consumer dissatisfaction, complaint distribution, and company responsiveness within the U.S. financial services industry. Using Tableau, Python, and Excel, the analysis reveals that the majority of complaints originate from credit reporting agencies—primarily Experian, Equifax, and TransUnion—and most commonly concern inaccurate information, misattributed accounts, and delays in investigations.

At the geographic level, Texas, Florida, and California report the highest number of complaints, reflecting both their large consumer populations and robust financial activity. More than 90% of complaints are submitted digitally, underscoring the public's strong preference for online engagement and accessibility. Operationally, company responsiveness is notably efficient, with an average response time of 0.29 days (approximately 7 hours) and a 99.8% on-time resolution rate. However, the prevalence of "Closed with explanation" outcomes highlights ongoing concerns about transparency and perceived fairness in complaint resolutions.

Overall, while companies demonstrate impressive timeliness and procedural efficiency, the persistence of data accuracy issues points to deeper structural challenges. Addressing these through improved data governance, clearer communication, and proactive monitoring will be essential to restoring consumer confidence and ensuring accountability across the financial ecosystem.

[1] *Source:* Consumer Financial Protection Bureau (2020). "CFPB Annual Complaint Report Highlights More Than a Half Million Complaints Received in 2020."

1. Introduction

The Consumer Complaint Database serves as a centralized repository of consumer grievances concerning financial products and services. Complaints are submitted through the Consumer Financial Protection Bureau's (CFPB) call center and subsequently forwarded to relevant companies for review and response. They are made public once the company confirms a commercial relationship with the consumer or after 15 days, whichever occurs first.

The dataset originally contained approximately 11.7 million records, providing an extensive foundation for exploring consumer sentiment and regulatory compliance trends. For this project, the dataset was cleaned and refined to a sample of 10,000 records, allowing for efficient exploration, visualization, and pattern detection without compromising representativeness. The primary objective of this analysis was to examine how consumer complaints vary across companies, identify key issues and product categories driving dissatisfaction, and assess how effectively firms respond to these concerns.

2. Tools and Methodology

The analysis employed a combination of data cleaning, visualization, and statistical techniques across three major tools:

- Data Cleaning & Sampling: Python and Microsoft Excel were used to preprocess data, handle missing values, filter irrelevant records, and generate a representative subset of 10,000 observations from the larger dataset.

- Visualization: Tableau dashboards provided interactive, geographic, and categorical visualizations that helped identify trends in complaint volume, issue type, and regional distribution.
- Statistical Analysis: Python was used to calculate descriptive statistics and analyze response time distributions, providing quantitative insights into company performance and complaint resolution efficiency.

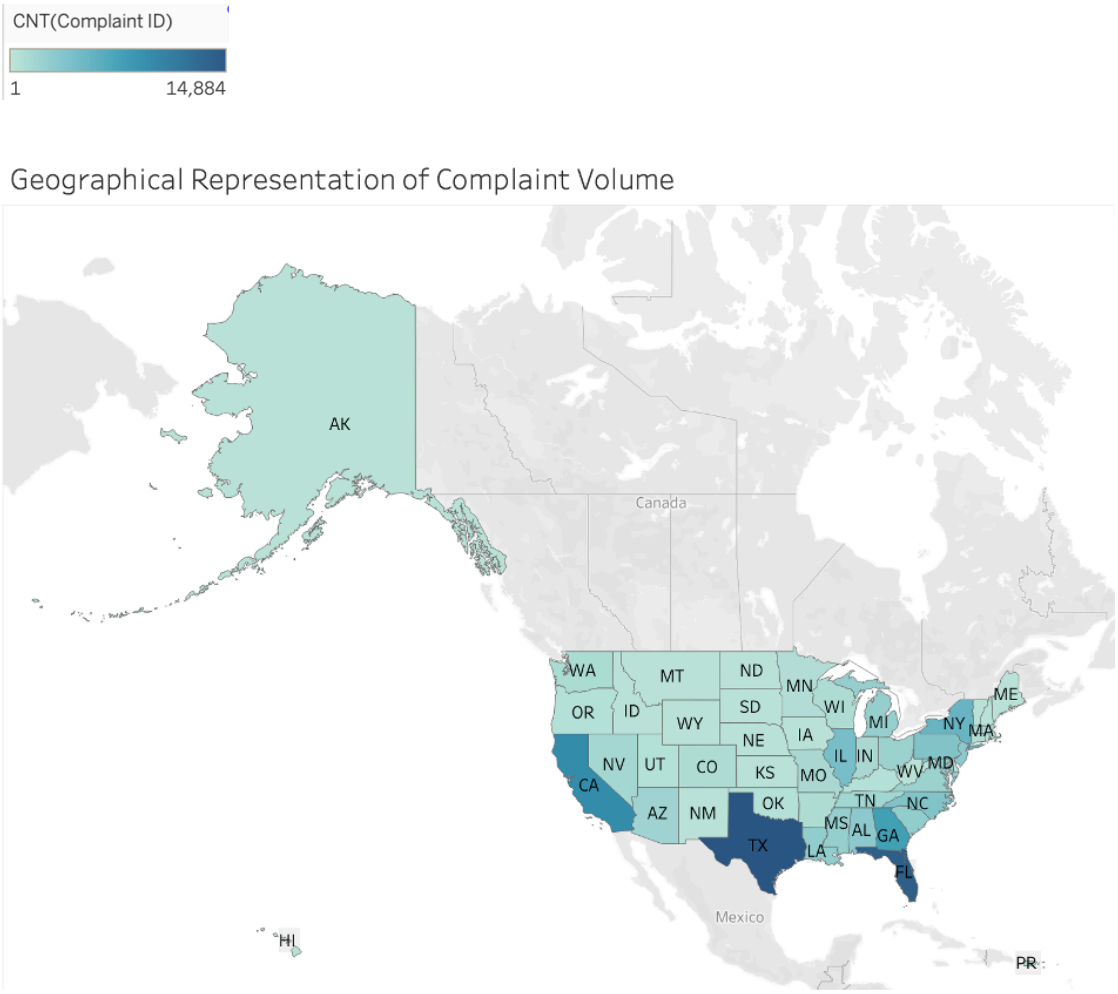
Together, these methods enabled a holistic understanding of consumer complaints, combining quantitative rigor with visual storytelling to highlight actionable insights for improving data accuracy, customer experience, and operational timeliness.

3. Tableau Visualizations

3.1 Complaints by State (Map Visualization)

This map illustrates the geographic distribution of consumer complaints across the United States. Key Findings:

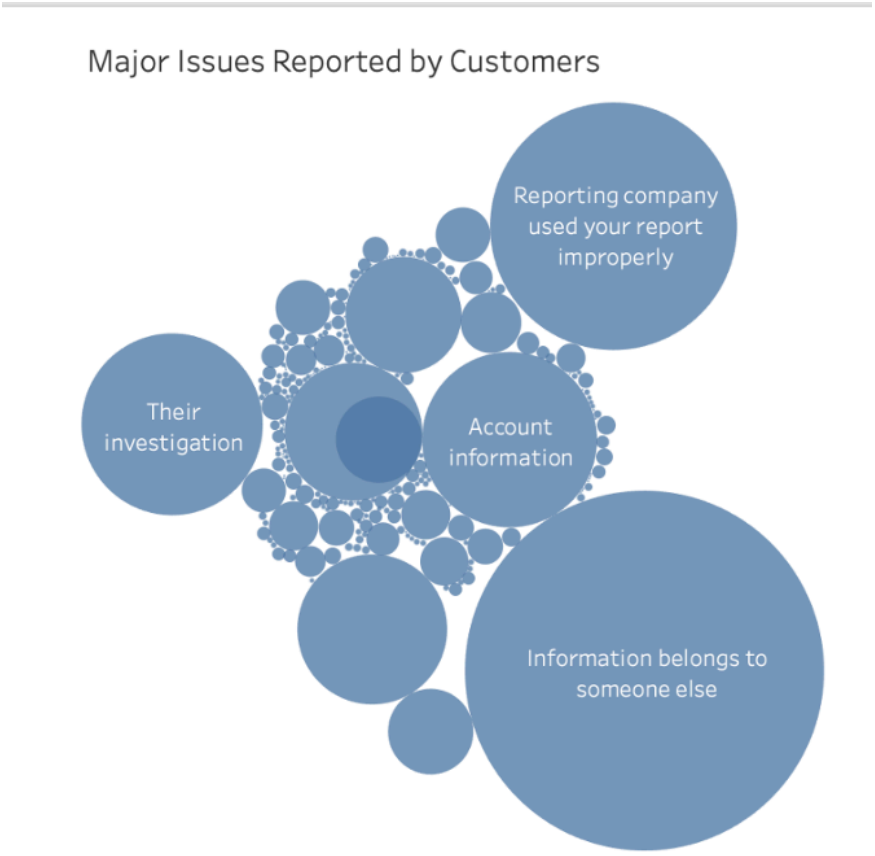
- Texas, Florida, and California report the highest complaint volumes.
- These states' elevated numbers correlate with larger populations and active financial markets.
- They represent critical focus areas for monitoring consumer dissatisfaction and regulatory oversight.



3.2 Common Issues (Bubble Chart)

The bubble chart displays the most frequently reported issues. Observations:

- Dominant issues include incorrect information on reports, accounts belonging to someone else, and investigation delays.
- These recurring complaints point to persistent data accuracy and verification issues in credit reporting agencies.



3.3 Complaint Submission Methods by Product (Heat Map)

This heat map compares submission methods across product categories.

Findings:

- Web submissions dominate across all categories, particularly for credit reporting and debt collection.
- Traditional methods such as fax and referral are nearly obsolete.

- The trend indicates a strong consumer preference for digital and immediate submission channels

Product Issue and Submission Method

Product	Submitted via				
	Fax	Phone	Postal m..	Referral	Web
Student loan	1	5	4	9	320
Prepaid card		15	1	1	101
Payday loan, title loan, personal l..		8	3	4	177
Payday loan, title loan, or person..					24
Payday loan					5
Mortgage	8	43	32	94	574
Money transfers					6
Money transfer, virtual currency,..		18	2	2	801
Debt or credit management		4	2		75
Debt collection	4	41	21	18	5,027
Credit reporting, credit repair se..	11	31	21	16	2,308
Credit reporting or other person..		74	70	25	86,180
Credit reporting	5	1	17	13	124
Credit card or prepaid card	3	22	5	23	208
Credit card		44	25	51	1,454
Consumer Loan		4		2	25
Checking or savings account	1	65	13	112	1,146

3.4 Company Response Timeliness (Bubble/Bar Chart)

The visualization examines company response times over the analyzed period.

Key Insights:

- Nearly all companies respond within 24 hours, with an average response time of less than a day.

"At-Risk" responses—those delayed or unresolved—account for less than 0.2% of total complaints, demonstrating robust operational efficiency and compliance.

4. Python Statistical Analysis

4.1 Distribution of Response Times and Timeliness

Python integration was used to generate histograms and descriptive statistics for response time.Summary Statistics:

- Mean: 0.29 days (~7 hours)
- Median: 0 days (same-day response)
- Standard Deviation: 2.79 days
- Maximum: 63 days (outliers representing complex cases)

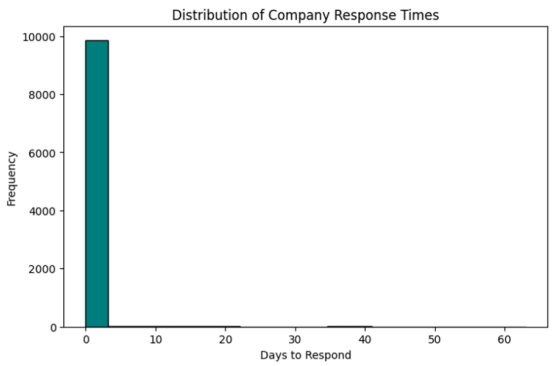
Interpretation:

The response time distribution is highly right-skewed, with a large concentration at 0 days.

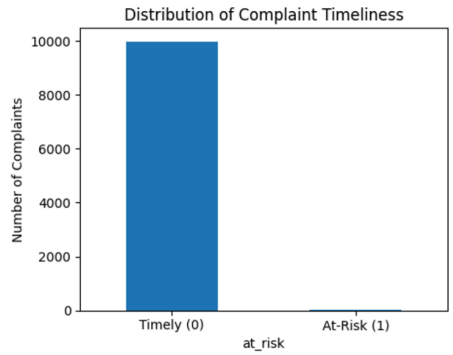
- 75% of complaints are handled the same day they are received.

- The histogram reveals one major spike at zero and a minimal long tail, confirming exceptional response consistency.

Complaint Timeliness Overview: These results reflect outstanding compliance and efficiency, with fewer than 2 delayed cases per 1,000 complaints. Companies appear to utilize automated or same-day processing systems to maintain timeliness.




```
Response Time Summary (in days):
count    10000.000000
mean         0.292700
std          2.791596
min          0.000000
25%          0.000000
50%          0.000000
75%          0.000000
max         63.000000
Name: days_to_respond, dtype: float64
```

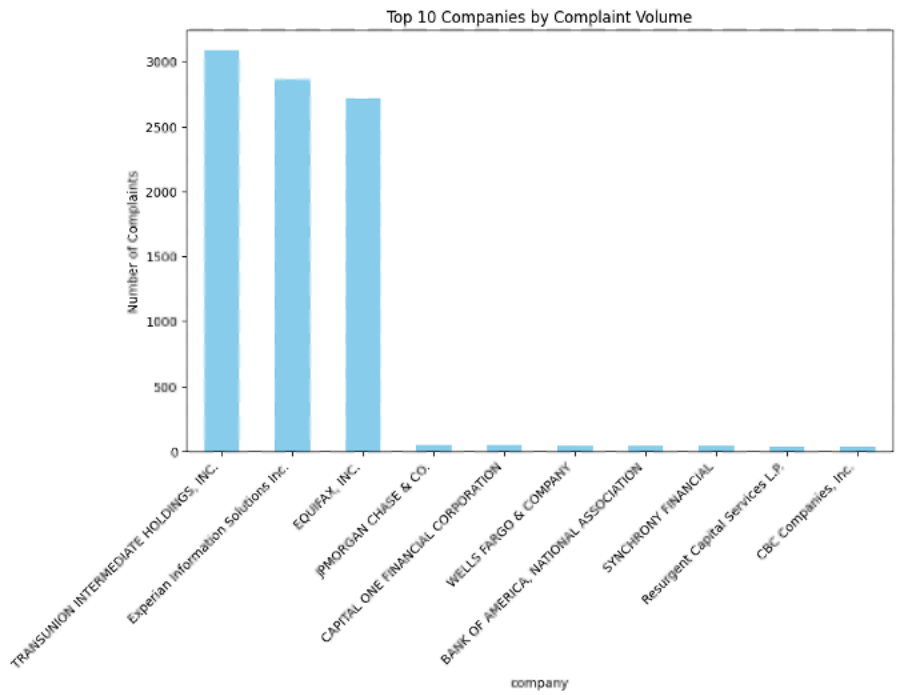


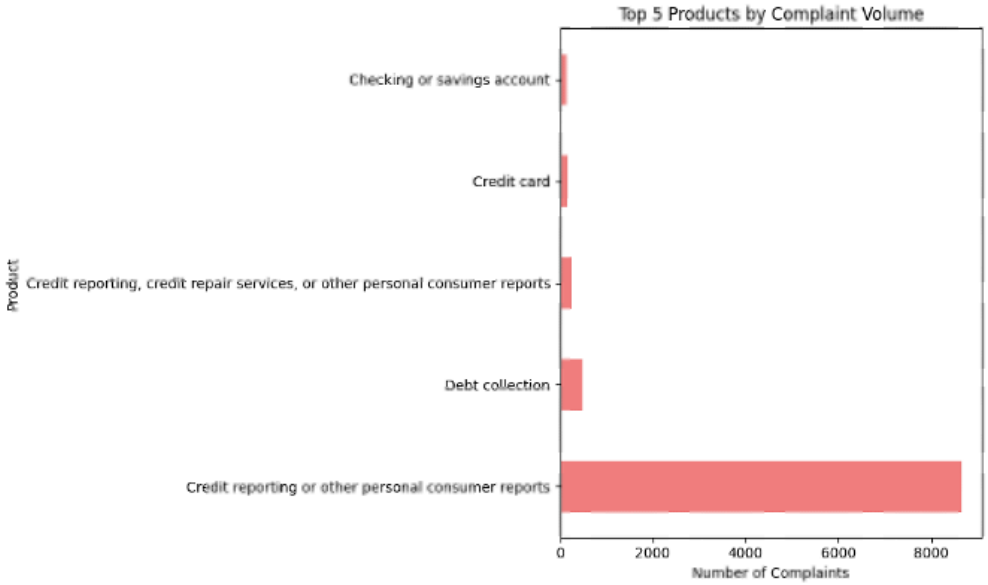

```
Timeliness Summary:
at_risk
Timely (0)     9981
At-Risk (1)      19
Name: count, dtype: int64

📈 Percentage Distribution:
at_risk
Timely (0)     99.81
At-Risk (1)     0.19
Name: proportion, dtype: float64
```

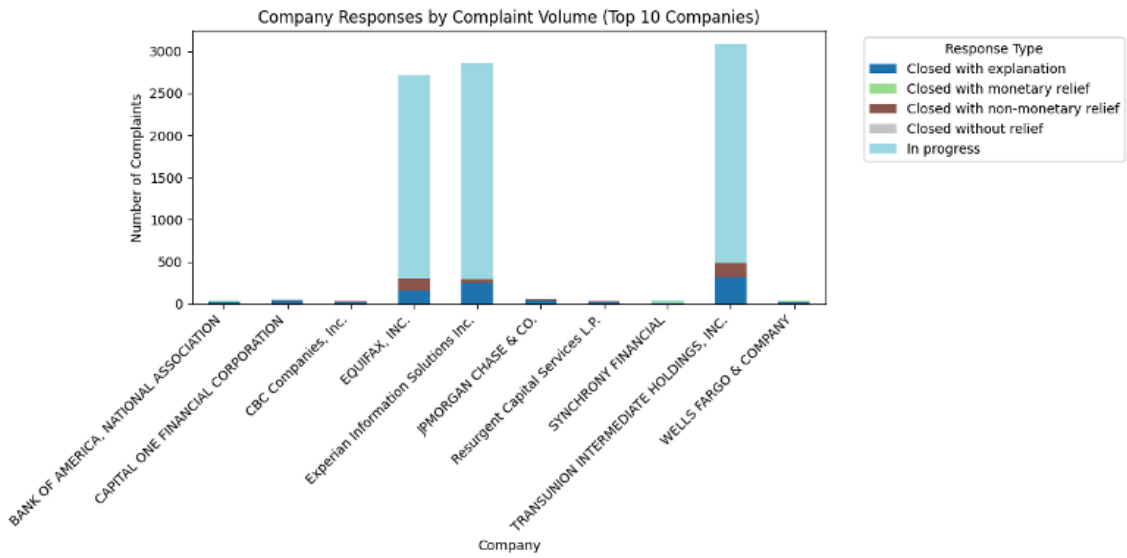
4.2 Top Companies and Product Dissatisfaction

The bar charts identify the companies and products most associated with consumer dissatisfaction.

- Experian, Equifax, and TransUnion dominate complaint volume and product dissatisfaction categories.
- The pattern highlights persistent consumer concerns about credit reporting accuracy and transparency, indicating a critical area for systemic improvement.


Top 10 Companies by Complaint Volume
Number of Complaints
3000
2500
2000
1500
1000
500
0
TRANSUNION INTERMEDIATE HOLDINGS, INC.
Experian Information Solutions Inc.
EQUIFAX, INC.
JPMORGAN CHASE & CO.
CAPITAL ONE FINANCIAL CORPORATION
WELLS FARGO & COMPANY
BANK OF AMERICA, NATIONAL ASSOCIATION
SYNCHRONY FINANCIAL
Resurgent Capital Services L.P.
CBC Companies, Inc.
company


Top 5 Products by Complaint Volume
Product
Checking or savings account
Credit card
Credit reporting, credit repair services, or other personal consumer reports
Debt collection
Credit reporting or other personal consumer reports
0
2000
4000
6000
8000
Number of Complaints



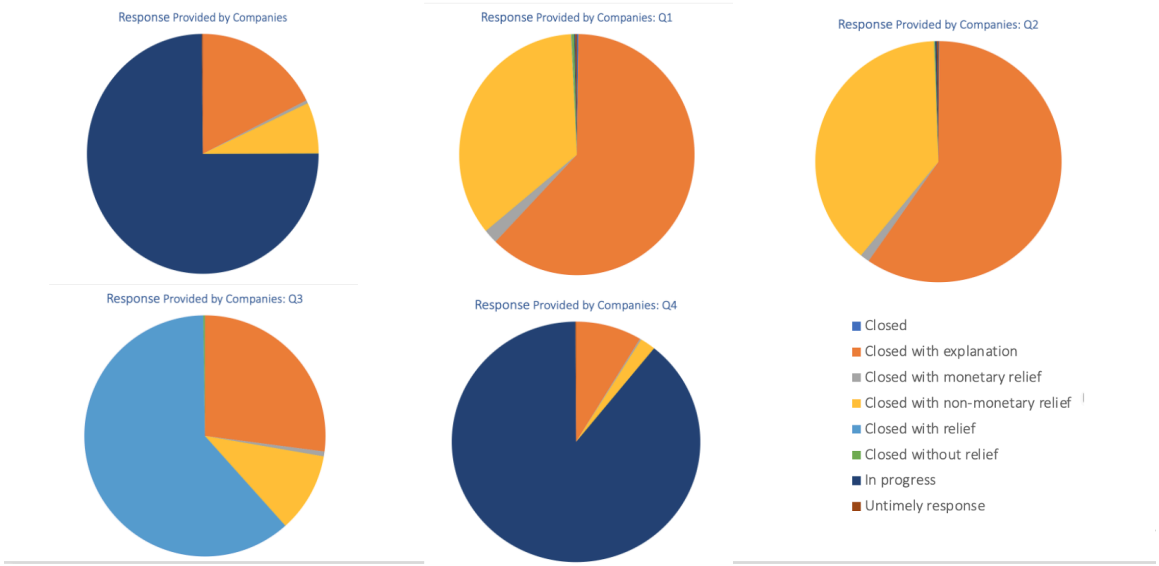
5. Quarterly Analysis (Excel Visuals)

5.1 Company Response Distribution

Stacked bar and pie charts depict how companies classify complaint resolutions.

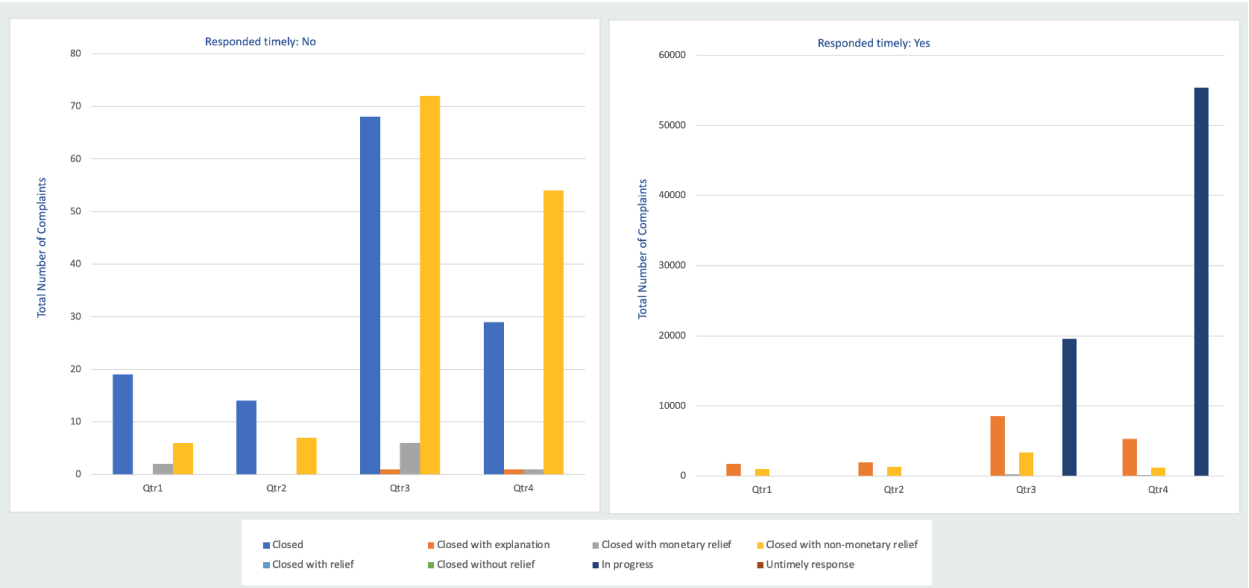
Insights:

- The most frequent outcome is “Closed with explanation,” followed by “Closed with monetary relief.”
- While most cases are resolved, the smaller proportion of monetary relief suggests potential consumer dissatisfaction with non-compensatory resolutions.



5.2 Quarterly Trends

- The proportion of resolution types remains stable across quarters.
- A minor surge in Q3 complaint volume does not significantly alter response patterns, indicating consistent operational strategies throughout the yea



Recommendations and Conclusion

To reduce recurring complaints and strengthen trust, companies must first focus on improving data accuracy through automated validation and regular reconciliation with credit bureaus. Greater transparency in case outcomes—such as clear resolution summaries and follow-up feedback—will ensure consumers feel heard and informed. Enhancing the digital complaint experience through real-time tracking and simplified forms can further improve satisfaction and efficiency.

High-complaint states like Texas, Florida, and California should be monitored using predictive dashboards that anticipate complaint surges, while automation in triaging and response systems should be maintained to preserve timely performance. Finally, firms should adopt preventive success measures—like first-contact resolution and repeat complaint tracking—and foster cross-industry data sharing to tackle systemic credit reporting issues.

In conclusion, while the industry demonstrates exceptional timeliness, lasting improvement depends on shifting from reactive compliance to proactive problem prevention, data integrity, and transparent consumer engagement.

Python Code Screenshots:

[ ]:
```python
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

[ ]:
```python
df = pd.read_csv("complaints.csv", nrows=1000)
```

[3]:
```python
print("Shape:", df.shape)
```

```
Shape: (1000, 18)
```

[8]:
```python
print(df.head())
```

```
  Date received                                            Product  \
0    2020-07-06  Credit reporting, credit repair services, or o...
1    2025-10-06  Credit reporting or other personal consumer re...
2    2025-10-14  Credit reporting or other personal consumer re...
3    2025-10-26  Credit reporting or other personal consumer re...
4    2025-10-10  Credit reporting or other personal consumer re...

        Sub-product                                          Issue  \
0  Credit reporting              Incorrect information on your report
1  Credit reporting  Problem with a company's investigation into an...
2  Credit reporting              Incorrect information on your report
3  Credit reporting              Incorrect information on your report
4  Credit reporting              Incorrect information on your report

                                          Sub-issue  \
0               Information belongs to someone else
1              Investigation took more than 30 days
2  Information is missing that should be on the r...
3               Information belongs to someone else
4               Information belongs to someone else

  Consumer complaint narrative  \
0                           NaN
1                           NaN
2                           NaN
3                           NaN
4                           NaN

                              Company public response  \
0  Company has responded to the consumer and the ...
1                                                NaN
2                                                NaN
3                                                NaN
4                                                NaN

                                 Company State ZIP code Tags  \
0     Experian Information Solutions Inc.    FL    346XX  NaN
1     Experian Information Solutions Inc.    TX    75454  NaN
2                          EQUIFAX, INC.    TX    75062  NaN
3  TRANSUNION INTERMEDIATE HOLDINGS, INC.    CA    93619  NaN
4                          EQUIFAX, INC.    GA    30341  NaN

  Consumer consent provided? Submitted via Date sent to company  \
0                      Other           Web           2020-07-06
1                        NaN           Web           2025-10-06
2                        NaN           Web           2025-10-14
3                        NaN           Web           2025-10-26
4                        NaN           Web           2025-10-10

  Company response to consumer Timely response? Consumer disputed?  \
0      Closed with explanation              Yes                NaN
1                  In progress              Yes                NaN
2                  In progress              Yes                NaN
3                  In progress              Yes                NaN
4                  In progress              Yes                NaN

   Complaint ID
0       3730948
1      16372503
2      16558024
3      16826857
4      16507707
```

[6]:
```python
print(df.columns.tolist())
```

```
['Date received', 'Product', 'Sub-product', 'Issue', 'Sub-issue', 'Consumer complaint narrative', 'Compan
'Complaint ID']
```

[13]:
```
df1 = pd.read_csv("complaints.csv")
```

[ ]:
```
print("Shape:", df1.shape)
```

[ ]:
```
print("Shape1:", df1.shape)
```

[ ]:
```
print("Number of rows:", len(df1))
```

[ ]:
```
df1.shape[0]
```

[ ]:
```
df.shape[0]
```

[ ]:
```
print(df.head())
```

[18]:
```
print("Shape:", df1.shape)
```

```
Shape: (11774995, 18)
```

[30]:
```
df1.columns = df1.columns.str.lower().str.replace(" ", "_").str.replace("-", "_")
```

[31]:
```
df1["state"] = df1["state"].fillna("Unknown")
```

[47]:
```
df1["timely_flag"] = df1["timely_response?"].str.lower().map({"yes": 1, "no": 0})
df1["at_risk"] = (df1["timely_flag"] == 0).astype(int)
```

[48]:
```
df1.head()
```

[48]:

	date_received	product	sub_product	issue	sub_issue	company	state	zip_c
0	2020-07-06	Credit reporting, credit repair services, or o...	Credit reporting	Incorrect information on your report	Information belongs to someone else	Experian Information Solutions Inc.	FL	346
1	2025-10-06	Credit reporting or other personal consumer re...	Credit reporting	Problem with a company's investigation into an...	Investigation took more than 30 days	Experian Information Solutions Inc.	TX	754
2	2025-10-14	Credit reporting or other personal consumer re...	Credit reporting	Incorrect information on your report	Information is missing that should be on the r...	EQUIFAX, INC.	TX	750
3	2025-10-26	Credit reporting or other personal consumer re...	Credit reporting	Incorrect information on your report	Information belongs to someone else	TRANSUNION INTERMEDIATE HOLDINGS, INC.	CA	93
4	2025-10-10	Credit reporting or other personal consumer re...	Credit reporting	Incorrect information on your report	Information belongs to someone else	EQUIFAX, INC.	GA	30

[60]:
```
df2 = pd.read_csv("complaints.csv",nrows=100000)
```

[61]:
```
df2.head()
```

[61]:

	Date received	Product	Sub-product	Issue	Sub-issue	Consumer complaint narrative	Company public response	Company	State	ZIP code	Tags	Consumer consent provided?	Submitted via	Date sent to company	Company response to consumer	Timely response?	Consumer disputed?	Comp
0	2020-07-06	Credit reporting, credit repair services, or o...	Credit reporting	Incorrect information on your report	Information belongs to someone else	NaN	Company has responded to the consumer and the ...	Experian Information Solutions Inc.	FL	346XX	NaN	Other	Web	2020-07-06	Closed with explanation	Yes	NaN	373
1	2025-10-06	Credit reporting or other personal consumer re...	Credit reporting	Problem with a company's investigation into an...	Investigation took more than 30 days	NaN	NaN	Experian Information Solutions Inc.	TX	75454	NaN	NaN	Web	2025-10-06	In progress	Yes	NaN	1637
2	2025-10-14	Credit reporting or other personal consumer re...	Credit reporting	Incorrect information on your report	Information is missing that should be on the r...	NaN	NaN	EQUIFAX, INC.	TX	75062	NaN	NaN	Web	2025-10-14	In progress	Yes	NaN	1655
3	2025-10-26	Credit reporting or other personal consumer re...	Credit reporting	Incorrect information on your report	Information belongs to someone else	NaN	NaN	TRANSUNION INTERMEDIATE HOLDINGS, INC.	CA	93619	NaN	NaN	Web	2025-10-26	In progress	Yes	NaN	1682
4	2025-10-10	Credit reporting or other personal consumer re...	Credit reporting	Incorrect information on your report	Information belongs to someone else	NaN	NaN	EQUIFAX, INC.	GA	30341	NaN	NaN	Web	2025-10-10	In progress	Yes	NaN	1650

[63]:
```
df2.to_csv("consumer_complaints_clean_100k.csv", index=False)
```

[64]:
```
print("✅ Clean file saved! Shape:", df2.shape)
```

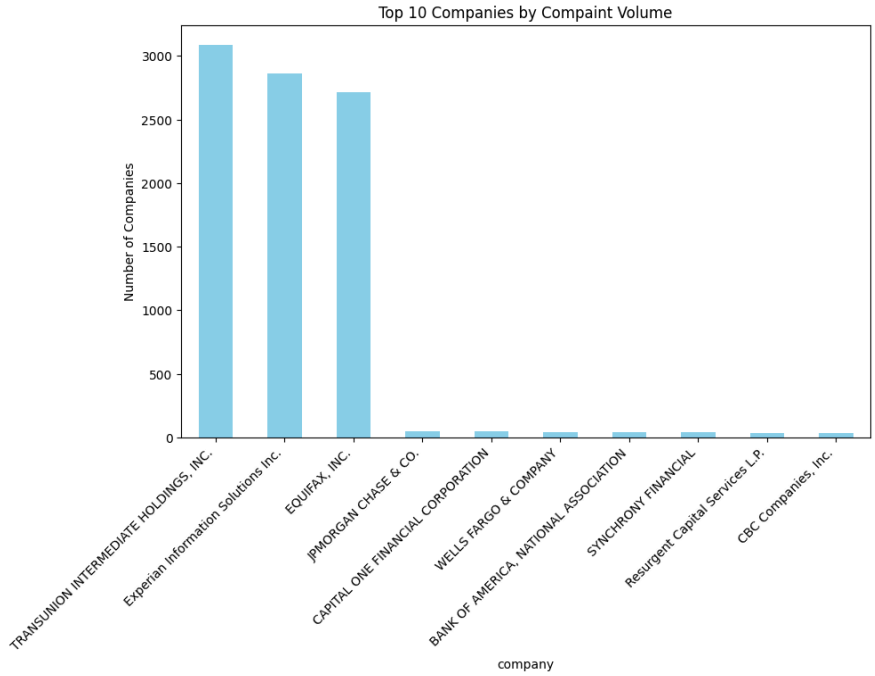
```
✅ Clean file saved! Shape: (100000, 18)
```

[71]:
```
top_companies = df1["company"].value_counts().head(10)
print(top_companies)
```

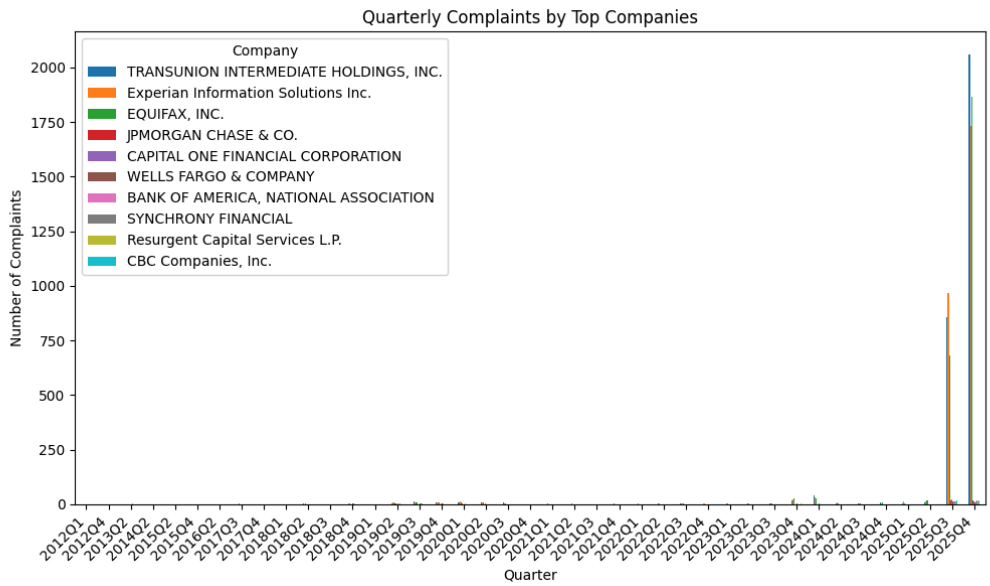
```
company
TRANSUNION INTERMEDIATE HOLDINGS, INC.    3088
Experian Information Solutions Inc.       2862
EQUIFAX, INC.                             2716
JPMORGAN CHASE & CO.                        51
CAPITAL ONE FINANCIAL CORPORATION           49
WELLS FARGO & COMPANY                       43
BANK OF AMERICA, NATIONAL ASSOCIATION       42
SYNCHRONY FINANCIAL                         39
Resurgent Capital Services L.P.             35
CBC Companies, Inc.                         35
Name: count, dtype: int64
```

[72]:
```
top_companies.plot(kind='bar', color='skyblue',figsize=(10,6))
plt.title('Top 10 Companies by Compaint Volume')
plt.ylabel('Number of Companies')
plt.xticks(rotation=45, ha='right')
plt.show
```

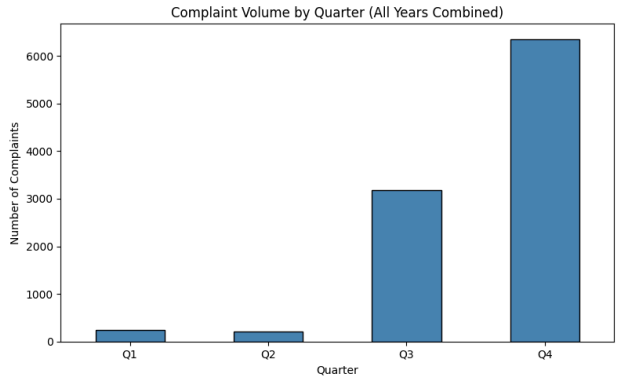
[72]:
```
<function matplotlib.pyplot.show(close=None, block=None)>
```



[73]:
```python
import pandas as pd
import matplotlib.pyplot as plt

df1['quarter'] = df1['date_received'].dt.to_period('Q')
grouped = df1.groupby(['quarter', 'company']).size().unstack().fillna(0)

top_companies = df1['company'].value_counts().head(10).index
grouped = grouped[top_companies]

grouped.plot(kind='bar', figsize=(10,6))
plt.title('Quarterly Complaints by Top Companies')
plt.ylabel('Number of Complaints')
plt.xlabel('Quarter')
plt.xticks(rotation=45, ha='right')
plt.legend(title='Company')
plt.tight_layout()
plt.show()
```



[74]:
```python
df1['date_received'] = pd.to_datetime(df1['date_received'], errors='coerce')

df1['quarter'] = df1['date_received'].dt.quarter

quarter_counts = df1['quarter'].value_counts().sort_index()

plt.figure(figsize=(8,5))
quarter_counts.plot(kind='bar', color='steelblue', edgecolor='black')

plt.title('Complaint Volume by Quarter (All Years Combined)')
plt.xlabel('Quarter')
plt.ylabel('Number of Complaints')
plt.xticks(ticks=[0,1,2,3], labels=['Q1','Q2','Q3','Q4'], rotation=0)
plt.tight_layout()
plt.show()
```



[75]:
```
top_product = df1["company"].value_counts().head(10)
print(top_product)
```

```
company
TRANSUNION INTERMEDIATE HOLDINGS, INC.    3088
Experian Information Solutions Inc.       2862
EQUIFAX, INC.                             2716
JPMORGAN CHASE & CO.                        51
CAPITAL ONE FINANCIAL CORPORATION           49
WELLS FARGO & COMPANY                       43
BANK OF AMERICA, NATIONAL ASSOCIATION       42
SYNCHRONY FINANCIAL                         39
Resurgent Capital Services L.P.             35
CBC Companies, Inc.                         35
Name: count, dtype: int64
```

[76]:
```
top_product = df1["product"].value_counts().head(10)
print(top_product)
```

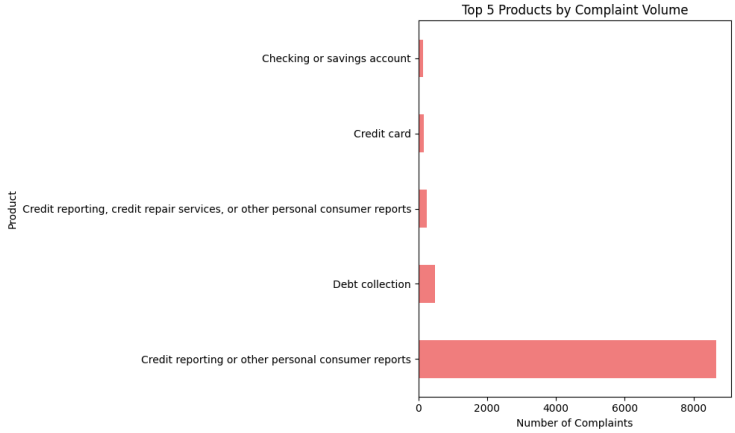
```
product
Credit reporting or other personal consumer reports                          8660
Debt collection                                                               485
Credit reporting, credit repair services, or other personal consumer reports  244
Credit card                                                                   149
Checking or savings account                                                   140
Money transfer, virtual currency, or money service                             85
Mortgage                                                                       77
Credit card or prepaid card                                                    42
Student loan                                                                   39
Vehicle loan or lease                                                          34
Name: count, dtype: int64
```

[77]:
```
top_product = df1["product"].value_counts().head(5)
print(top_product)
```

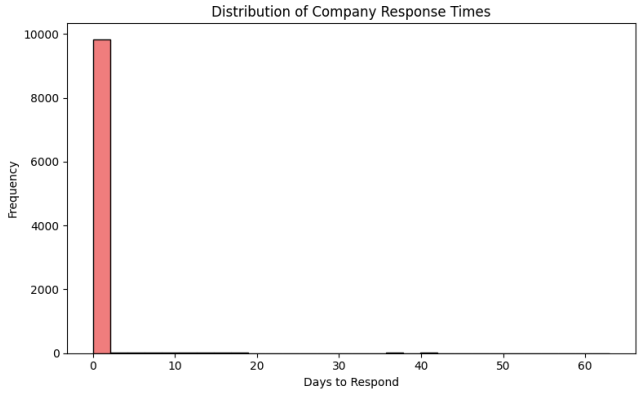
```
product
Credit reporting or other personal consumer reports                          8660
Debt collection                                                               485
Credit reporting, credit repair services, or other personal consumer reports  244
Credit card                                                                   149
Checking or savings account                                                   140
Name: count, dtype: int64
```

[78]:
```
top_product.plot(kind='bar', color='pink',figsize=(10,6))
plt.title('Top 5 Product by Compaint Volume')
plt.ylabel('Number of Complaints')
plt.xticks(rotation=45, ha='right')
plt.show
```

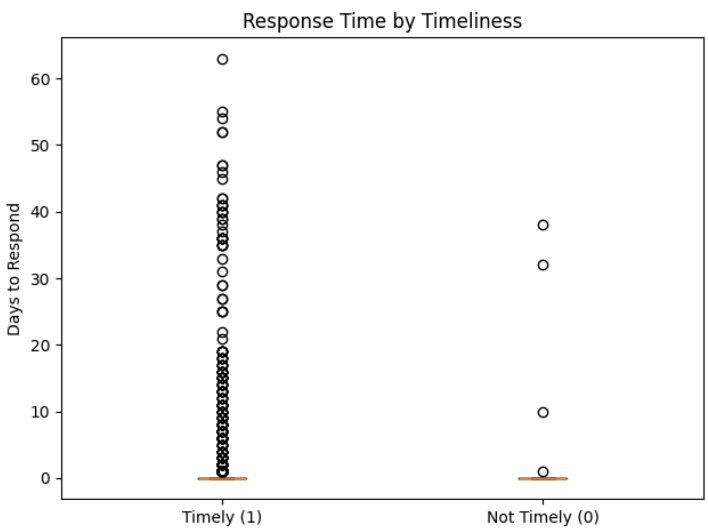
[80]:
```
top_products = df1['product'].value_counts().head(5)
plt.figure(figsize=(10,6))
top_products.plot(kind='barh', color='lightcoral')
plt.title('Top 5 Products by Complaint Volume')
plt.xlabel('Number of Complaints')
plt.ylabel('Product')
plt.tight_layout()
plt.show()
```



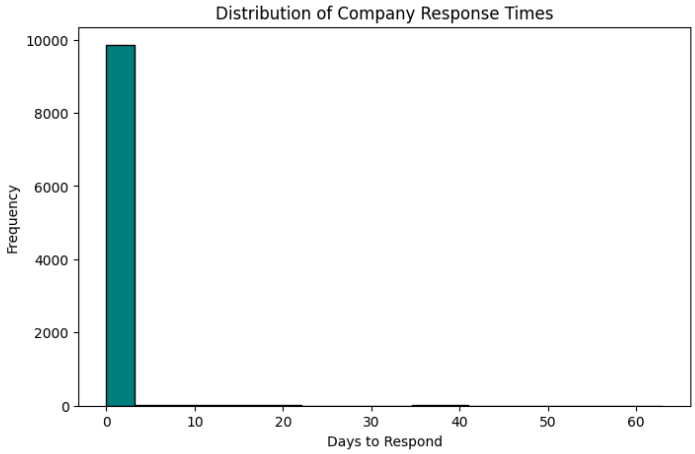
[96]:
```
df1['days_to_respond'].plot.hist(
    bins=30, color='lightcoral', edgecolor='black', figsize=(8,5))
plt.title('Distribution of Company Response Times')
plt.xlabel('Days to Respond')
plt.ylabel('Frequency')
plt.tight_layout()
plt.show()
```



[97]:
```
import matplotlib.pyplot as plt

groups = [df1.loc[df1['timely_flag']==1, 'days_to_respond'],
          df1.loc[df1['timely_flag']==0, 'days_to_respond']]
plt.boxplot(groups, labels=['Timely (1)', 'Not Timely (0)'])
plt.title('Response Time by Timeliness')
plt.ylabel('Days to Respond')
plt.tight_layout()
plt.show()
```

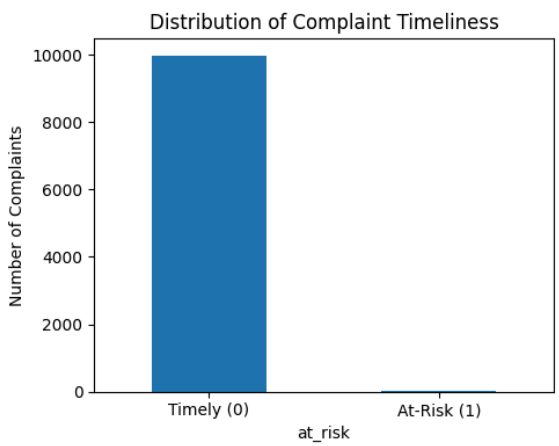


[99]:
```
plt.figure(figsize=(8,5))
plt.hist(df['days_to_respond'], bins=20, color='teal', edgecolor='black')
plt.title('Distribution of Company Response Times')
plt.xlabel('Days to Respond')
plt.ylabel('Frequency')
plt.show()
```

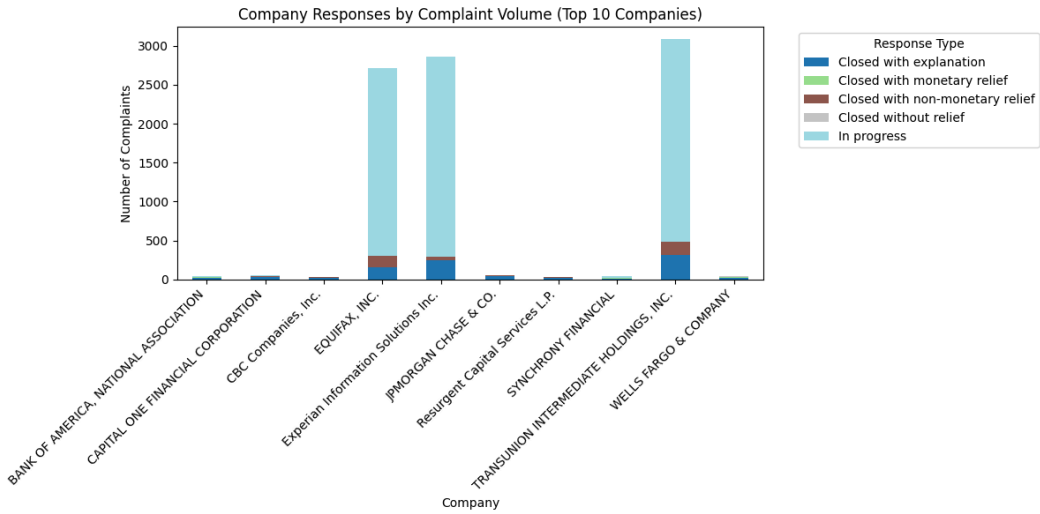


[103]:
```python
import matplotlib.pyplot as plt
import pandas as pd

# 1) Make sure we're using the right df
# (If you want to keep using `df`, alias it)
df = df1.copy()

# 2) Normalize column names once to avoid case/space mismatches
df.columns = df.columns.str.strip().str.lower().str.replace(' ', '_')

# 3) (Re)create timely_flag and at_risk if missing
if 'timely_flag' not in df.columns:
    df['timely_flag'] = (
        df['timely_response?'].astype(str).str.lower().map({'yes': 1, 'no': 0})
    )

if 'at_risk' not in df.columns:
    df['at_risk'] = (df['timely_flag'] == 0).astype(int)

# 4) Quick sanity check
print(df[['timely_response?', 'timely_flag', 'at_risk']].head())

# 5) Plot counts (Matplotlib only)
counts = df['at_risk'].value_counts().reindex([0, 1], fill_value=0)
plt.figure(figsize=(5,4))
counts.plot(kind='bar')
plt.xticks([0,1], ['Timely (0)', 'At-Risk (1)'], rotation=0)
plt.title('Distribution of Complaint Timeliness')
plt.ylabel('Number of Complaints')
plt.tight_layout()
plt.show()
```

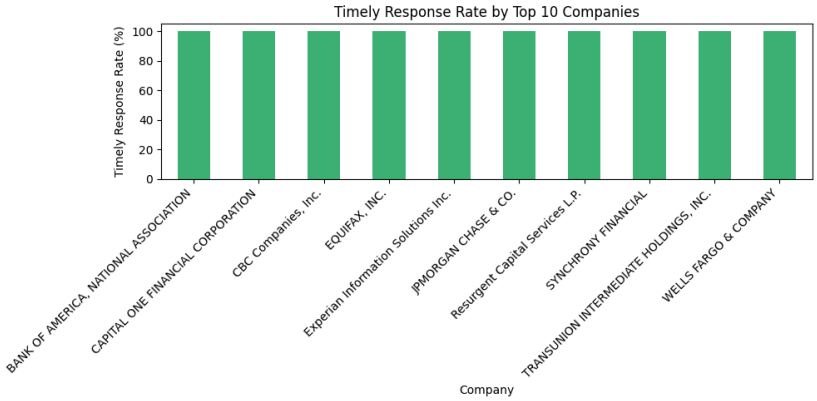
```
  timely_response?  timely_flag  at_risk
0              Yes            1        0
1              Yes            1        0
2              Yes            1        0
3              Yes            1        0
4              Yes            1        0
```



[109]:
```python
import matplotlib.pyplot as plt
import pandas as pd

top_companies = df1['company'].value_counts().head(10).index

subset = df1[df1['company'].isin(top_companies)]

pivot = pd.crosstab(subset['company'], subset['company_response_to_consumer'])


pivot.plot(kind='bar', stacked=True, figsize=(12,6), colormap='tab20')
plt.title('Company Responses by Complaint Volume (Top 10 Companies)')
plt.xlabel('Company')
plt.ylabel('Number of Complaints')
plt.xticks(rotation=45, ha='right')
plt.legend(title='Response Type', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()
```

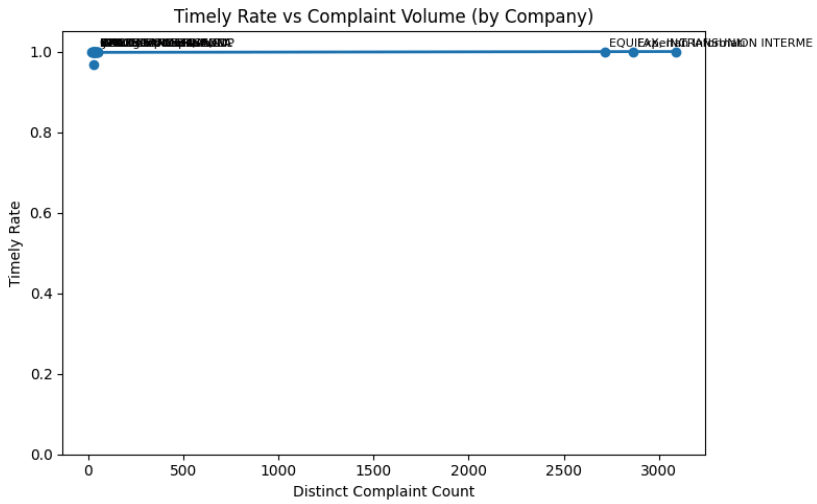


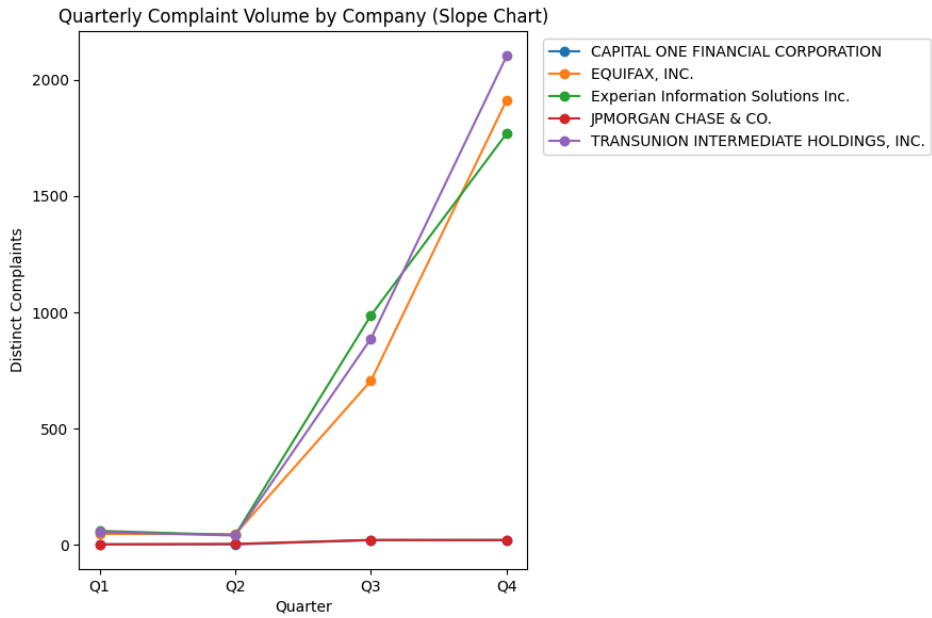
[110]:
```python
timely_summary = (
    df1[df1['company'].isin(top_companies)]
    .groupby('company')['timely_flag']
    .mean() * 100
).sort_values(ascending=False)

timely_summary.plot(kind='bar', color='mediumseagreen', figsize=(10,5))
plt.title('Timely Response Rate by Top 10 Companies')
plt.xlabel('Company')
plt.ylabel('Timely Response Rate (%)')
plt.xticks(rotation=45, ha='right')
plt.ylim(0, 105)
plt.tight_layout()
plt.show()
```



[112]:

```
grp = df1.groupby('company').agg(
    complaint_count=('complaint_id','nunique'),
    timely_rate=('timely_flag','mean')
).reset_index()

grp = grp.sort_values('complaint_count', ascending=False).head(20)

x = grp['complaint_count'].values.astype(float)
y = grp['timely_rate'].values.astype(float)

coef = np.polyfit(x, y, 1)
poly1d_fn = np.poly1d(coef)

plt.figure(figsize=(8,5))
plt.scatter(x, y)
plt.plot(np.sort(x), poly1d_fn(np.sort(x)), linewidth=2)  # trendline
plt.title('Timely Rate vs Complaint Volume (by Company)')
plt.xlabel('Distinct Complaint Count'); plt.ylabel('Timely Rate')
plt.ylim(0, 1.05)
for _, r in grp.head(10).iterrows():
    plt.annotate(r['company'][:18], (r['complaint_count'], r['timely_rate']), fontsize=8, xytext=(3,3), textcoords='offset points')
plt.tight_layout(); plt.show()
```



[113]:

```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

df1['quarter'] = pd.to_datetime(df1['date_received'], errors='coerce').dt.quarter
top_companies = df1['company'].value_counts().head(5).index
subset = df1[df1['company'].isin(top_companies)]

agg = (subset.groupby(['company','quarter'])['complaint_id']
              .nunique().reset_index(name='count'))

wide = agg.pivot(index='company', columns='quarter', values='count').fillna(0)

plt.figure(figsize=(9,6))
x = [1,2,3,4]
for comp, row in wide.iterrows():
    y = [row.get(q, 0) for q in [1,2,3,4]]
    plt.plot(x, y, marker='o', label=comp)

plt.xticks([1,2,3,4], ['Q1','Q2','Q3','Q4'])
plt.title('Quarterly Complaint Volume by Company (Slope Chart)')
plt.xlabel('Quarter'); plt.ylabel('Distinct Complaints')
plt.legend(bbox_to_anchor=(1.02, 1), loc='upper left')
plt.tight_layout(); plt.show()
```